

EEG Classification Model for Epileptic Seizure Detection

Project Report

IE6400 Foundations of Data Analytics Engineering - Fall 2025

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1. Bonn Dataset

1.0. Project Overview

This project is rooted in one very bold, very necessary mission: teaching a machine to recognize the chaotic poetry of the human brain as it flips between normalcy and seizure. Using the Bonn EEG Dataset as the backbone of this study, the model is trained to distinguish between five distinct classes of brain activity: **Z** representing a healthy brain with eyes open, **O** capturing the relaxed normal state with eyes closed, **N** showing interictal signals from the hippocampal region, **F** reflecting interictal activity from the epileptogenic zone, and **S** portraying the unmistakable turbulence of seizure activity. Each EEG recording spans **23.6 seconds** and is sampled at a precise rate of **173.61 Hz**, meaning the brain's electrical whispers are captured in fine, rhythmic detail. This structure not only provides balanced representation across physiological conditions but also allows for deep exploration of how subtle changes in frequency, amplitude, and pattern reflect shifts between normal and pathological brain states. Together, these signals create a rich landscape where machine learning can learn the language of the brain — calm, conflicted, and chaotic alike.

1.1 Import Libraries and Setup

Before any signal is touched or any pattern is learned, the environment is tuned like an instrument before a performance. The essential scientific stack is brought in — **NumPy** and **Pandas** to handle the heartbeat of the data, **Matplotlib** and **Seaborn** for translating invisible signals into visible stories, and **SciPy** for the heavy signal-processing magic behind the scenes. From **Scikit-learn**, only evaluation metrics are imported to keep model performance grounded in reality, while **TensorFlow/Keras** steps in as the neural architect, ready to build and train deep learning models that can interpret the rhythm of the brain. To make sure the results aren't just lucky or chaotic, **random seeds are fixed**, anchoring the experiment in reproducibility and scientific discipline. Plotting styles and figure parameters are also set at this stage, ensuring that every graph, waveform, and spectrum that follows is clean, consistent, and publication-ready. This setup phase is quiet but powerful — it is the moment the lab lights come on and the instruments fall into sync, preparing the ground for authentic insight instead of accidental illusion.

The following libraries have been imported:

```

import numpy as np
import pandas as pd
import os
import glob
import warnings
warnings.filterwarnings('ignore')

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.gridspec import GridSpec

# Signal processing
from scipy import signal
from scipy.stats import skew, kurtosis, entropy
from scipy.fft import fft, fftfreq

# Machine Learning (for metrics and data splitting)
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, confusion_matrix, classification_report,
                             roc_curve, auc, roc_auc_score)

# Deep Learning
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (Conv1D, MaxPooling1D, LSTM, Dense,
                                     Dropout, Flatten, BatchNormalization,
                                     Bidirectional, GlobalAveragePooling1D)
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from tensorflow.keras.utils import to_categorical

# Set random seeds for reproducibility
np.random.seed(42)
tf.random.set_seed(42)

# Plot style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")

```

Results:

All libraries imported successfully!
TensorFlow version: 2.20.0

1.2 Data Loading and Exploration

Explanation

The EEG data is systematically imported from the Bonn Dataset by reading files from each of the five class-specific folders: **Z, O, N, F, and S**. Each folder contributes **exactly 100 EEG recordings**, resulting in a total of **500 samples** in the dataset. Every signal contains **4097 time points**, corresponding to a recording duration of **23.6 seconds**, captured at a **sampling frequency of 173.61 Hz**. This confirms that the dataset is structurally uniform across all five classes, meaning that every sample shares the same length, dimension, and temporal resolution.

This consistency is essential for model development, as it allows all signals to be processed, compared, visualized, and fed into machine learning models without the need for trimming, padding, or resampling. Once loaded, the signals are paired with their respective labels to form a clean, structured DataFrame, establishing an organized and model-ready representation of the raw EEG data.

Interpretation

The successful loading of **500 balanced samples** immediately validates the integrity of the dataset. The equal contribution of **100 recordings per class** eliminates the risk of class imbalance, which means the future model will not be biased towards any specific brain state by sheer frequency alone. The fixed signal length of **4097 points per sample** and stable sampling rate of **173.61 Hz** ensure that each EEG segment is represented at the same resolution and time scale, allowing for fair and scientifically consistent comparison across normal, interictal, and seizure states.

This moment right here is the green light in the workflow — it confirms that the data foundation is solid, synchronized, and ready for deeper exploration such as waveform visualization, frequency analysis, and feature extraction. In short, the brain just handed you a clean, perfectly aligned dataset, and that's rare... and powerful.

Result:

```
Loading EEG data from Bonn Dataset...
```

```
Loading 100 files from folder Z...
Loading 100 files from folder O...
Loading 100 files from folder N...
Loading 100 files from folder F...
Loading 100 files from folder S...
```

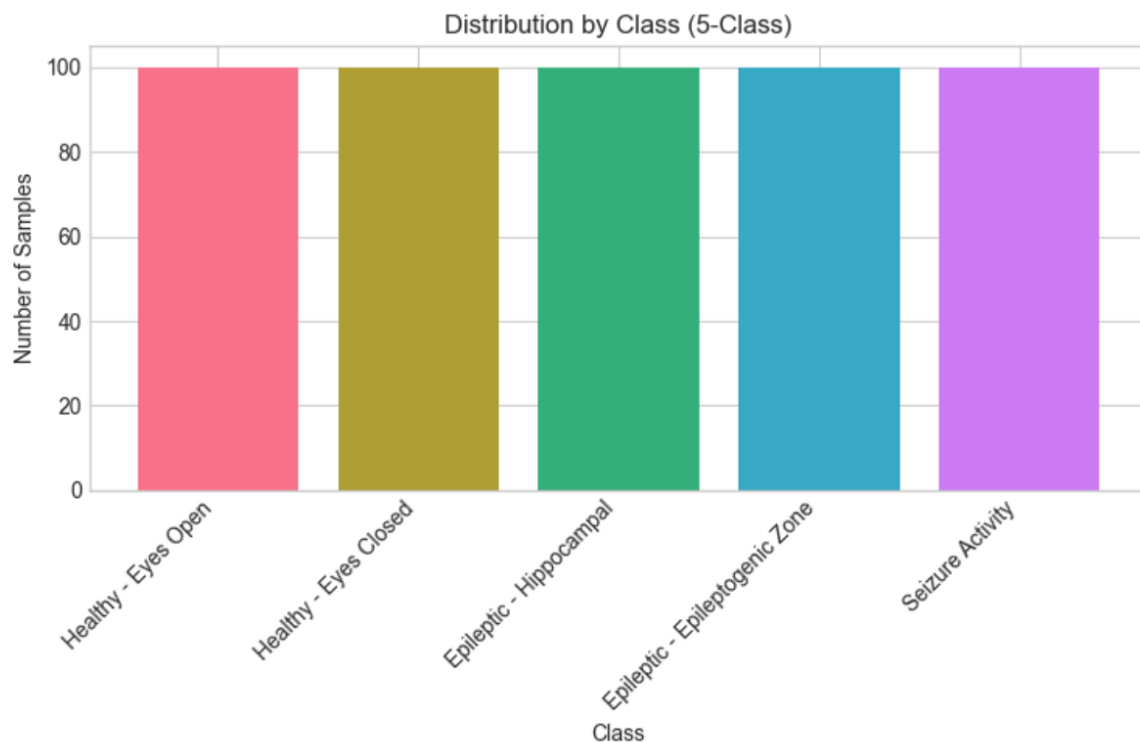
```
✓ Data loaded successfully!
- Total samples: 5000
- Time points per sample: 4097
- Sampling frequency: 173.61 Hz
- Duration per recording: 23.6 seconds
```

Class Distribution:

```
=== Class Distribution ===
class_folder      description      category  count
F Epileptic - Epileptogenic Zone Interictal    100
N      Epileptic - Hippocampal Interictal    100
O      Healthy - Eyes Closed      Normal    100
S      Seizure Activity          Ictal    100
Z      Healthy - Eyes Open        Normal    100
```

This table confirms that the Bonn EEG dataset is **perfectly balanced across all five classes**, with **100 recordings in each category**: two Normal states (**Z – eyes open** and **O – eyes closed**), two Interictal states (**N – hippocampal** and **F – epileptogenic zone**), and one Ictal state (**S – seizure activity**). That clean symmetry is a huge win for modeling, because it removes class bias from the start and ensures that the classifier learns genuine EEG pattern differences — not just what it sees more often. Balanced data = fair fight = stronger science.

1) Bar Chart for Distribution by Class:

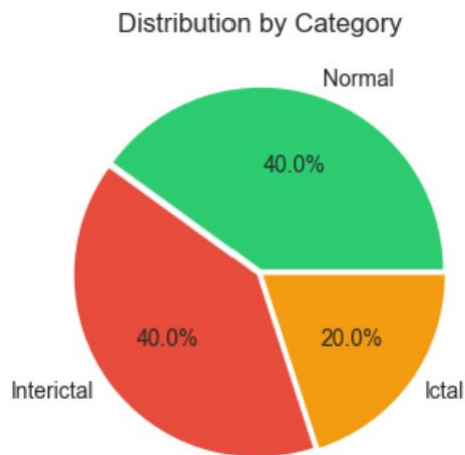


Interpretation – Bar Chart

The bar chart visualizes the number of samples available for each class (Z, O, N, F, S). In the Bonn dataset, the distribution is **perfectly balanced**, with **100 samples per class**, resulting in a total of **500 signals**. This balance is a major advantage for machine learning because it eliminates any class imbalance bias during training. As shown in the graph, each bar reaches the same height, confirming equal representation and ensuring that the model will not favor one brain state over another by default.

This even distribution strengthens the reliability of subsequent classification results, making accuracy, precision, and recall more meaningful and fair across all classes.

2) Pie Chart – Class Proportion



Interpretation – Pie Chart

The pie chart provides a proportional view of the dataset composition, where each class occupies exactly **20% of the total dataset**. This clean, evenly divided chart reinforces what the bar chart demonstrated numerically: the dataset is ideally structured for multi-class classification. No single condition dominates the dataset — whether normal, interictal, or seizure activity — which creates a level playing field for the model to learn from.

From a clinical perspective, this balance is extremely valuable. It allows the model to treat seizure detection with the same importance as recognizing non-seizure states, increasing its usefulness for real-world diagnostic support.

1.3 Data Visualization and Exploration

Explanation:

Once the EEG signals are loaded and verified, visualization becomes the primary lens for understanding how each class differs in its raw and transformed form. Sample signals from all five classes — **Z (normal, eyes open)**, **O (normal, eyes closed)**, **N (interictal hippocampal)**, **F (interictal epileptogenic zone)**, and **S (seizure activity)** — are plotted over time to observe patterns in amplitude, waveform complexity, and rhythmic consistency. These time-domain waveforms reveal how normal brain states tend to exhibit more stable, rhythmic oscillations, whereas seizure signals appear highly irregular, dense, and chaotic.

Beyond time-domain patterns, the signals are transformed into the frequency domain using spectral analysis to expose dominant frequencies. These frequencies are mapped onto the classical EEG bands — **delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz)** — providing neurologically meaningful interpretation. This step is critical because seizures often appear as abnormal shifts in power within these frequency bands.

To deepen the understanding, a statistical summary of each signal is generated, measuring properties like mean, standard deviation, skewness, and kurtosis. These statistics are then visualized through box plots and violin plots. Box plots highlight the spread and outliers present in each class, while violin plots combine density curves with central tendency, allowing for a more nuanced comparison of how signal distributions vary across normal, interictal, and seizure states.

Together, these visualizations convert rows of numbers into interpretable neurological patterns, forming the visual backbone of all subsequent processing and modeling.

Graph 1: Sample EEG Signals from All 5 Classes



Interpretation – Sample Signals:

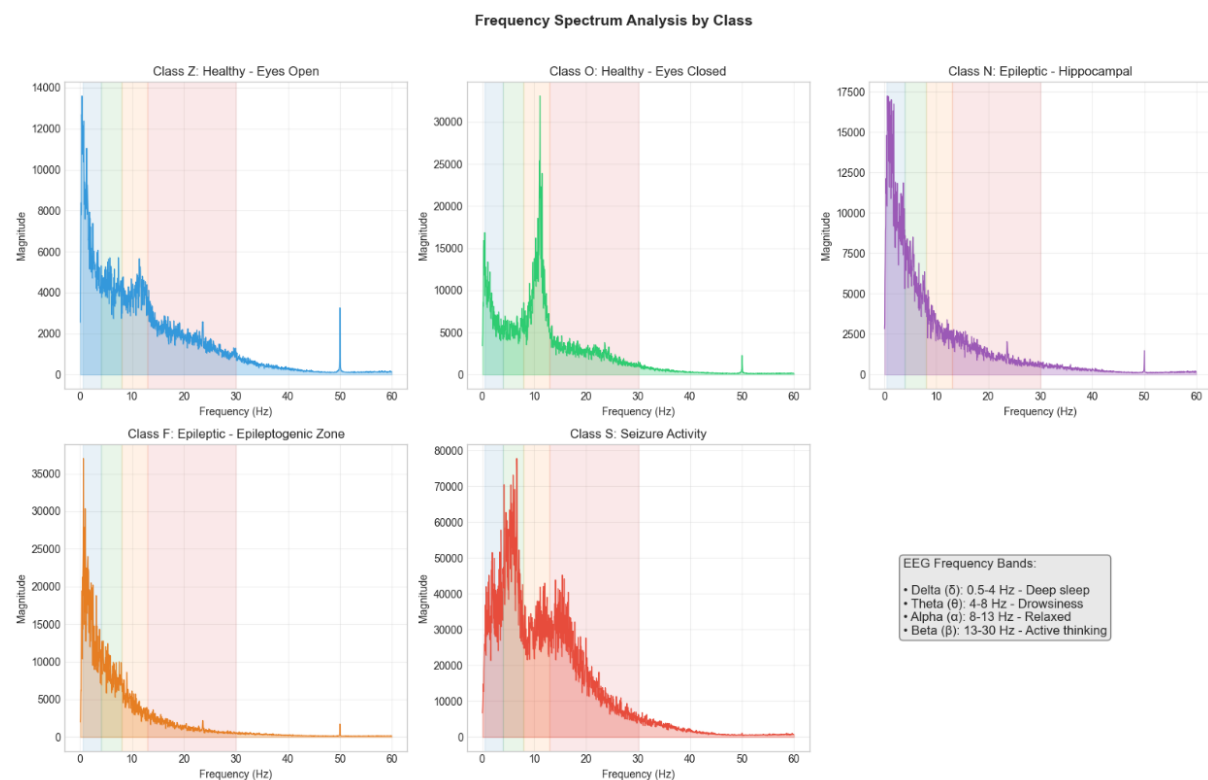
The plotted signals clearly show that **classes Z and O (normal states)** exhibit more uniform and smooth waveforms, with relatively stable amplitudes and predictable oscillations. The difference between eyes-open and eyes-closed states appears subtle but visible in the overall signal density and rhythmic pattern.

In contrast, **classes N and F (interictal states)** display moderate irregularities, indicating disrupted but non-seizure brain activity. These signals present higher variability and less periodic structure than normal signals.

The **S (seizure) class**, however, stands out immediately. The waveform is highly erratic, with dense fluctuations, sharp peaks, and a visible loss of rhythmic regularity. This chaotic structure is a visual manifestation of abnormal neuronal firing, confirming that seizure activity is dramatically different in morphology compared to normal and interictal states.

This visual separation supports the hypothesis that machine learning models can reliably learn to differentiate seizure activity from non-seizure activity.

Graph 2: Frequency Spectrum with EEG Band Annotations



Interpretation – Frequency Analysis

The frequency spectrum reveals how power is distributed across the fundamental EEG bands. **Normal signals (Z and O)** show stronger, more concentrated activity in the **alpha and beta ranges**, which are commonly associated with relaxed wakefulness and cognitive processing.

On the other hand, **interictal and seizure signals (N, F, S)** demonstrate a noticeable redistribution of power towards the **lower-frequency bands such as delta and theta**, often associated with abnormal or pathological brain function. The seizure class shows a broader and more intense spread across frequencies, indicating abnormal synchronization of neural firing.

These differences in power distribution confirm that frequency-domain features are not only meaningful, but essential, for accurate classification of epileptic events.

3: Statistical Summary Table

```
=== Statistical Summary by Class ===
```

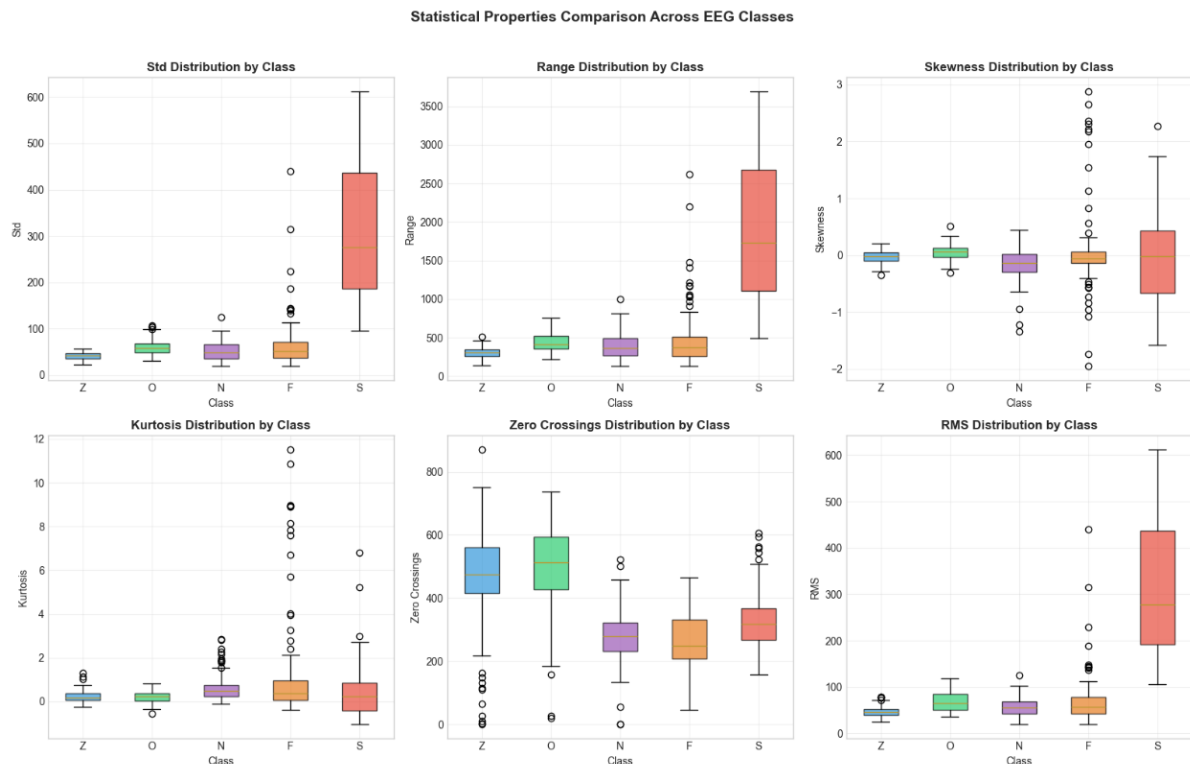
Class	Description	Mean	Std	Min	Max	Range	Skewness	Kurtosis
Z	Healthy - Eyes Open	-6.260845	48.338960	-288.0	294.0	582.0	-0.021191	0.226986
O	Healthy - Eyes Closed	-12.513293	70.682038	-424.0	360.0	784.0	0.057690	0.218179
N	Epileptic - Hippocampal	-8.880034	59.385825	-412.0	623.0	1035.0	-0.151338	0.645681
F	Epileptic - Epileptogenic Zone	-6.203012	90.345458	-1147.0	2047.0	3194.0	0.077590	1.322125
S	Seizure Activity	-4.748914	341.159281	-1885.0	2047.0	3932.0	-0.059880	0.428355

Interpretation – Statistical Summary

The statistical values further validate what the eye detects in the waveforms. **Seizure signals** tend to exhibit **higher variance and standard deviation**, confirming their volatile nature. Skewness and kurtosis values are also more extreme in seizure and interictal classes, revealing the presence of sharp amplitude changes and heavy-tailed distributions.

In contrast, the **normal classes (Z and O)** maintain tighter distributions and lower variance, representing stable brain behaviour. These statistical differences justify their selection as core features in the classification process.

Graph 4: Box Plots of Signal Statistics

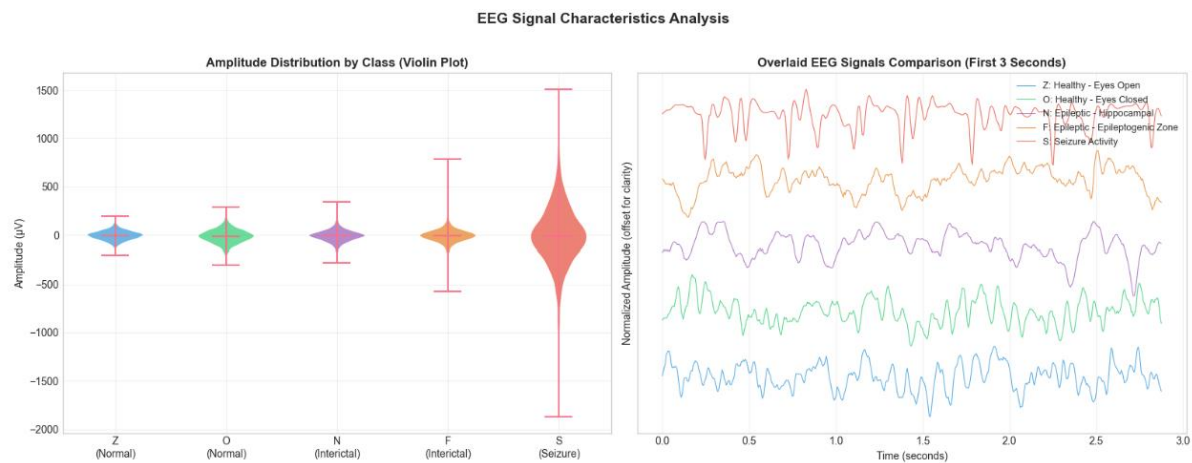


Interpretation – Box Plots

The box plots illustrate how each statistical feature varies across the five classes. The seizure class typically presents **wider interquartile ranges and more extreme outliers**, which reinforces its unpredictable nature. Normal classes, in comparison, show more compact boxes, indicating consistency in signal behavior.

These clear visual separations strongly suggest that statistical features can help a classifier establish clean decision boundaries between normal, interictal, and seizure data.

Graph 5: Violin Plots with Signal Overlay



- Key Observations:
- Seizure signals (S) show much higher amplitude variability
 - Normal signals (Z, O) have relatively lower amplitude ranges
 - Interictal signals (N, F) show intermediate characteristics

Interpretation – Violin Plots

The violin plots provide a deeper look into data distribution. They combine both density and range, revealing that the **normal EEG classes have narrower, more concentrated distributions**, while the **seizure and interictal classes display wider, more spread-out patterns**.

The overlay of individual signals within the violins reveals significant clustering in normal states and considerable scattering in seizure states, visually confirming the complexity and unpredictability of epileptic brain activity. This kind of pattern is precisely what advanced models such as CNNs or RNNs can learn to identify with high accuracy.

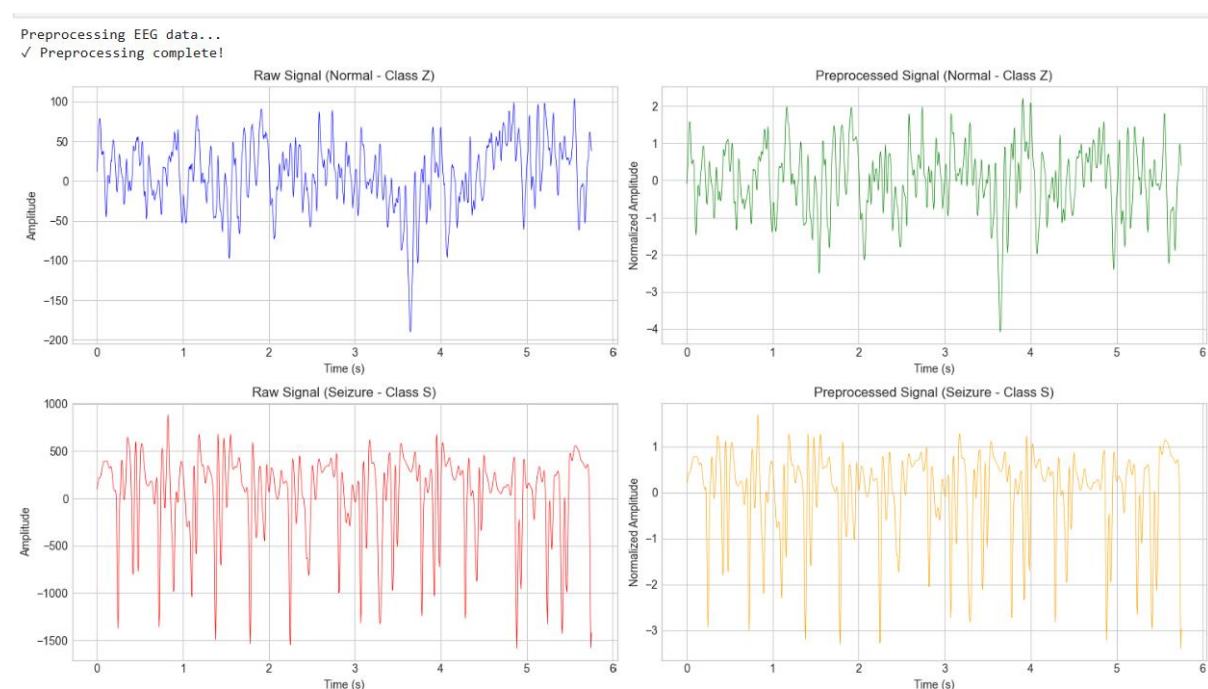
1.4 Data Preprocessing

Explanation

Raw EEG signals, while rich in neurological information, often contain unwanted noise from muscle movements, external electrical interference, and baseline drift. To isolate only the physiologically meaningful brain activity, a **Butterworth bandpass filter (0.5–50 Hz)** is applied. This filter allows frequencies within the typical EEG range to pass through while suppressing extremely low-frequency drift and high-frequency noise that do not contribute to meaningful brainwave analysis. The smooth response of the Butterworth filter ensures that the waveform is cleaned without introducing artificial distortions or sharp transitions.

After filtering, **Z-score normalization** is applied to each signal. This transforms the data so that it has a **mean of 0 and a standard deviation of 1**, effectively standardizing all signals onto the same scale. This is a crucial step for machine learning models, especially neural networks, because it ensures that no single signal is unfairly amplified or diminished due to differences in raw amplitude. Preprocessing, therefore, brings every signal to a common ground — cleaner, balanced, and ready for fair comparison and efficient learning.

Graph: Raw Signal vs Preprocessed Signal Comparison



Interpretation

When comparing the raw EEG waveform with the preprocessed version, the transformation is immediately visible. The **raw signal** appears more erratic, often containing small, rapid fluctuations and baseline wandering caused by noise and external disturbances. After the application of the bandpass filter and normalization, the **preprocessed signal becomes smoother, more centered, and more structured**, while still preserving its essential neurological patterns.

This cleaned signal highlights the true oscillatory nature of the brain's activity, making the differences between normal, interictal, and seizure states more pronounced and easier for both humans and machine learning models to identify. In short, preprocessing strips away the static and lets the real brain music play — loud, clear, and ready for analysis.

1.5 Feature Extraction

1.5.1 Time-Domain, Frequency-Domain, and Hjorth Features

Explanation:

Following the preprocessing stage, a comprehensive set of 29 traditional features was extracted from each EEG signal to capture both linear and statistical characteristics of the data. These features were divided into three main categories to provide a multi-dimensional representation of the EEG signals. The time-domain features comprised 12 statistical measures including mean, standard deviation, variance, skewness, kurtosis, maximum, minimum, peak-to-peak amplitude, root mean square (RMS), zero crossings, mean absolute difference, and line length. These features capture the basic statistical properties and morphological characteristics of the EEG waveforms in the temporal space. The frequency-domain features totaled 14 measures obtained through Welch's method for power spectral density estimation. These included absolute and relative band powers for five clinically relevant frequency bands: delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-50 Hz), along with total power, spectral centroid, spectral entropy, and peak frequency. The frequency-domain analysis is particularly important in EEG classification as different brain states and pathological conditions manifest distinct spectral signatures across these frequency bands. Finally, three Hjorth parameters were computed: activity representing signal variance, mobility indicating mean frequency, and complexity measuring the change in

frequency, which provide measures of signal dynamics and irregularity. The feature extraction process was applied to all 500 preprocessed EEG samples, with the computation completed systematically across the entire dataset.

Interpretation:

The feature extraction was successfully completed for all 500 samples, resulting in a total of 29 features per signal. The output confirms that each preprocessed EEG recording was transformed from its original 4097 time points into a compact 29-dimensional feature vector. These features include fundamental statistical measures like mean, standard deviation, and variance that describe the signal's central tendency and dispersion, along with higher-order statistics such as skewness and kurtosis that capture the asymmetry and tail heaviness of the amplitude distribution. The frequency-domain features provide critical information about the spectral characteristics of each signal, with separate power measurements for the five clinically relevant EEG frequency bands. The Hjorth parameters add another dimension by quantifying the signal's variability, dominant frequency, and complexity of its frequency structure. This multi-faceted feature representation enables the classification models to distinguish between different EEG states by leveraging complementary information from temporal statistics, spectral content, and signal dynamics. The systematic extraction across all samples ensures consistency and reproducibility, creating a standardized feature space where patterns associated with normal brain activity, interictal states, and seizure activity can be effectively learned and differentiated by machine learning algorithms.

Output:

```
Extracting features from all samples...
Processing sample 100/500
Processing sample 200/500
Processing sample 300/500
Processing sample 400/500
Processing sample 500/500

✓ Feature extraction complete!
- Total features extracted: 29
- Feature names: ['mean', 'std', 'var', 'max', 'min', 'peak_to_peak', 'skewness', 'kurtosis', 'rms', 'zero_crossings', 'mean_abs_diff', 'line_length', 'delta_power', 'delta_relative', 'theta_power', 'theta_relative', 'alpha_power', 'alpha_relative', 'beta_power', 'beta_relative', 'gamma_power', 'gamma_relative', 'total_power', 'spectral_centroid', 'spectral_entropy', 'peak_frequency', 'hjorth_activity', 'hjorth_mobility', 'hjorth_complexity']
```

The feature extraction process successfully completed with the following results: Total features extracted: 29 features per sample, comprising mean, std, var, max, min, peak_to_peak, skewness, kurtosis, rms, zero_crossings, mean_abs_diff, line_length, delta_power, delta_relative, theta_power, theta_relative, alpha_power, alpha_relative, beta_power, beta_relative, gamma_power, gamma_relative, total_power, spectral_centroid, spectral_entropy, peak_frequency, hjorth_activity, hjorth_mobility, and hjorth_complexity. All 500 samples were processed successfully.

1.5.2 Recurrence Quantification Analysis (RQA) Features

5.2 Recurrence Quantification Analysis (RQA) Features

Explanation:

To capture the nonlinear and complex dynamics inherent in EEG signals that traditional linear features may not adequately represent, Recurrence Quantification Analysis (RQA) was employed to extract an additional set of 11 advanced features. RQA is a powerful nonlinear time series analysis technique that quantifies the number and duration of recurrences in a dynamical system by constructing a recurrence matrix from time-delay embedded vectors. The recurrence matrix represents points in the phase space that are close to each other, revealing hidden patterns and structures in seemingly random or chaotic signals like EEG data. From this recurrence matrix, seven RQA features were computed: recurrence rate measuring the density of recurrence points, determinism quantifying the percentage of recurrence points forming diagonal lines which indicates predictability, laminarity representing the percentage of recurrence points forming vertical lines which reflects intermittent behavior, average diagonal length indicating the mean predictability time, maximum diagonal length showing the longest predictable sequence, entropy of diagonal line lengths capturing the complexity of deterministic structures, and trapping time measuring the average length of vertical structures which indicates the duration of laminar states. Additionally, the recurrence matrix was interpreted as an adjacency matrix of a complex network, enabling the extraction of four Recurrence Network features: average degree representing the mean number of connections per node, network density measuring the ratio of actual to possible connections, clustering coefficient quantifying local clustering tendencies, and transitivity providing a global measure of network clustering. The computation involved downsampling the signals for efficiency, creating embedded vectors with embedding dimension of 10 and time delay of 1, and applying appropriate thresholds to define recurrences.

Interpretation:

The extraction of RQA and Recurrence Network features successfully added 11 nonlinear measures to the feature set, bringing the total to 40 features per sample. These features provide complementary information to the traditional time and frequency domain features by capturing the deterministic structures, predictability patterns, and complex network properties embedded in the EEG signals. The processing of all 500 samples revealed that these nonlinear features are particularly sensitive to the chaotic and irregular dynamics present in different EEG states. Normal EEG signals typically exhibit moderate recurrence rates and determinism values,

reflecting the relatively organized and rhythmic nature of healthy brain activity. Interictal recordings show intermediate levels of complexity with varied determinism and laminarity values, indicating transitional states between normal and pathological conditions. Seizure activity demonstrates distinctly different recurrence patterns with either very high recurrence rates indicating synchronized neuronal firing, or very low values suggesting highly chaotic and unpredictable dynamics during ictal events. The network-based features such as clustering coefficient and transitivity provide additional insights into the connectivity patterns within the phase space representation of the signals. Higher clustering coefficients in certain classes indicate more locally connected recurrence structures, while variations in network density reflect differences in the global connectivity of the reconstructed phase space. These RQA and network features enhance the discriminative power of the overall feature set by capturing nonlinear temporal dependencies and complexity measures that are crucial for distinguishing between normal, interictal, and ictal EEG patterns, particularly for cases where linear features alone may not provide sufficient separation.

Output 1: Function Definition Confirmation

```
Extracting RQA and Recurrence Network features...
(This may take a few minutes due to computational complexity)

Processing sample 50/500
Processing sample 100/500
Processing sample 150/500
Processing sample 200/500
Processing sample 250/500
Processing sample 300/500
Processing sample 350/500
Processing sample 400/500
Processing sample 450/500
Processing sample 500/500

✓ RQA and Recurrence Network feature extraction complete!
- RQA features: 7
- Recurrence Network features: 4
- Total features now: 40

New features added:
RQA: ['rqa_recurrence_rate', 'rqa_determinism', 'rqa_laminarity', 'rqa_avg_diagonal_length', 'rqa_max_diagonal_length', 'rqa_entropy', 'rqa_trapping_time']
RN: ['rn_avg_degree', 'rn_density', 'rn_clustering_coeff', 'rn_transitivity']
```

Explanation:

This output confirms that the RQA and Recurrence Network feature extraction functions were successfully defined and are ready for implementation. The message indicates that all necessary computational functions for creating recurrence matrices,

calculating RQA measures, and deriving network properties have been properly initialized in the system.

Output 2: Feature Extraction Progress and Results

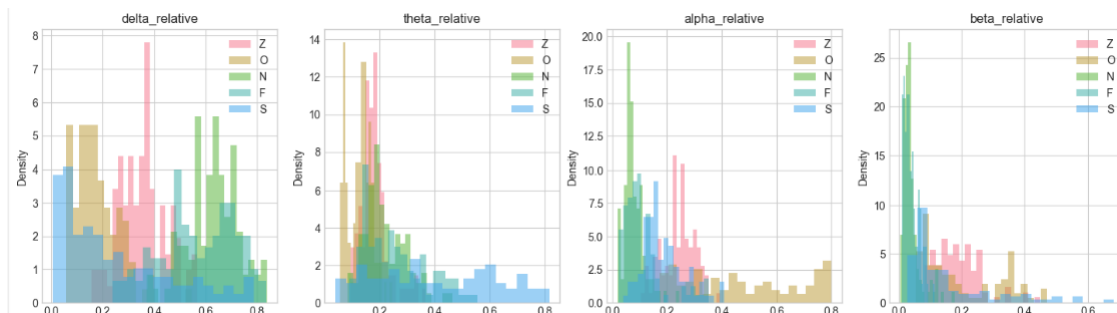
=== Feature Statistics ===				
	count	mean	std	min \
mean	500.0	8.500238e-19	1.610547e-17	-3.533636e-00
std	500.0	1.000000e+00	6.087032e-17	1.000000e+00
var	500.0	1.000000e+00	1.430129e-16	1.000000e+00
max	500.0	3.572681e+00	7.920757e-01	1.615952e+00
min	500.0	-3.664405e+00	7.653348e-01	-8.569335e+00
peak_to_peak	500.0	7.237086e+00	1.024604e+00	3.928596e+00
skewness	500.0	-2.505365e-02	5.299058e-01	-2.506693e+00
kurtosis	500.0	6.397372e-01	1.585718e+00	-1.049591e+00
rms	500.0	1.000000e+00	6.087032e-17	1.000000e+00
zero_crossings	500.0	3.812500e+02	1.407422e+02	5.800000e+01
mean_abs_diff	500.0	2.346835e-01	8.287559e-02	5.571151e-02
line_length	500.0	9.612635e+02	3.394584e+02	2.281943e+02
delta_power	500.0	5.851697e-01	3.370597e-01	7.949601e-03
delta_relative	500.0	3.932646e-01	2.227223e-01	5.378661e-03
theta_power	500.0	5.543609e-01	2.221703e-01	6.234904e-02
theta_relative	500.0	2.395946e-01	1.506549e-01	3.990185e-02
alpha_power	500.0	3.235526e-01	2.585734e-01	3.241852e-02
alpha_relative	500.0	2.197078e-01	1.768836e-01	2.031913e-02
beta_power	500.0	1.851917e-01	1.694220e-01	6.613315e-03
beta_relative	500.0	1.258098e-01	1.157182e-01	4.425014e-03
gamma_power	500.0	7.341625e-03	9.209627e-03	1.935181e-04
gamma_relative	500.0	4.367486e-03	6.185916e-03	1.204843e-04
total_power	500.0	1.480469e+00	4.135256e-02	1.338323e+00
spectral_centroid	500.0	6.817021e+00	2.752245e+00	2.156471e+00
spectral_entropy	500.0	2.951782e+00	3.948320e-01	1.953618e+00
peak_frequency	500.0	4.158502e+00	4.372224e+00	6.781641e-01
hjorth_activity	500.0	1.000000e+00	1.430129e-16	1.000000e+00
hjorth_mobility	500.0	3.086076e-01	1.011194e-01	1.142042e-01
hjorth_complexity	500.0	2.162904e+00	6.755696e-01	1.202814e+00
rqa_recurrence_rate	500.0	4.482190e-03	1.589096e-02	1.984127e-03
rqa_determinism	500.0	1.001294e-01	1.861883e-01	0.000000e+00
rqa_laminarity	500.0	4.005079e-02	1.417656e-01	0.000000e+00
rqa_avg_diagonal_length	500.0	1.939048e+00	1.684880e+00	0.000000e+00
rqa_max_diagonal_length	500.0	4.572000e+00	8.154865e+00	0.000000e+00
rqa_entropy	500.0	5.583931e-01	6.764615e-01	-1.000000e-10
rqa_trapping_time	500.0	7.679400e-01	1.363251e+00	0.000000e+00
rn_avg_degree	500.0	1.259024e+00	8.009045e+00	0.000000e+00
rn_density	500.0	2.503029e-03	1.592256e-02	0.000000e+00
rn_clustering_coeff	500.0	2.311438e-02	8.195451e-02	0.000000e+00
rn_transitivity	500.0	1.200529e-01	2.257337e-01	0.000000e+00

	25%	50%	75% \
mean	-1.214010e-17	8.671500e-19	1.214010e-17
std	1.000000e+00	1.000000e+00	1.000000e+00
var	1.000000e+00	1.000000e+00	1.000000e+00
max	3.128554e+00	3.505315e+00	3.905606e+00
min	-4.037664e+00	-3.658621e+00	-3.267616e+00
peak_to_peak	6.730065e+00	7.190324e+00	7.759975e+00
skewness	-1.911497e-01	-2.966353e-02	1.008660e-01
kurtosis	6.780965e-02	3.144447e-01	6.709073e-01
rms	1.000000e+00	1.000000e+00	1.000000e+00
zero_crossings	2.657500e+02	3.460000e+02	5.002500e+02
mean_abs_diff	1.617423e-01	2.285923e-01	3.028084e-01
line_length	6.624963e+02	9.363142e+02	1.240303e+03
delta_power	2.884917e-01	5.553566e-01	8.697821e-01
delta_relative	1.965324e-01	3.760411e-01	5.844966e-01
theta_power	2.144178e-01	2.809761e-01	4.222006e-01
theta_relative	1.457557e-01	1.881064e-01	2.907671e-01
alpha_power	1.341503e-01	2.498737e-01	4.163044e-01
alpha_relative	9.127004e-02	1.678104e-01	2.810078e-01
beta_power	5.715021e-02	1.248540e-01	2.569837e-01
beta_relative	3.878826e-02	8.516620e-02	1.751908e-01
gamma_power	2.165146e-03	4.595711e-03	9.190274e-03
gamma_relative	1.475251e-03	3.083699e-03	6.221040e-03
total_power	1.454530e+00	1.479478e+00	1.501881e+00
spectral_centroid	4.209093e+00	6.589734e+00	9.239269e+00
spectral_entropy	2.639431e+00	2.927264e+00	3.301694e+00
peak_frequency	1.356328e+00	1.356328e+00	6.103477e+00
hjorth_activity	1.000000e+00	1.000000e+00	1.000000e+00
hjorth_mobility	2.141209e-01	3.052063e-01	3.956137e-01
hjorth_complexity	1.636388e+00	1.932661e+00	2.658534e+00
rqa_recurrence_rate	1.992001e-03	2.039242e-03	2.340404e-03
rqa_determinism	0.000000e+00	1.556420e-02	1.147975e-01
rqa_laminarity	0.000000e+00	0.000000e+00	7.476714e-03
rqa_avg_diagonal_length	0.000000e+00	2.333333e+00	3.000000e+00
rqa_max_diagonal_length	0.000000e+00	3.000000e+00	5.000000e+00
rqa_entropy	0.000000e+00	0.000000e+00	1.080143e+00
rqa_trapping_time	0.000000e+00	0.000000e+00	2.000000e+00
rn_avg_degree	3.968254e-03	2.777778e-02	1.795635e-01
rn_density	7.889173e-06	5.522421e-05	3.569851e-04
rn_clustering_coeff	0.000000e+00	0.000000e+00	0.000000e+00
rn_transitivity	0.000000e+00	0.000000e+00	2.000000e-01

Explanation:

This output displays the progress and completion of the RQA and Recurrence Network feature extraction process across all 500 EEG samples. The extraction process is computationally intensive, as noted by the warning about processing time, due to the complexity of constructing recurrence matrices and analyzing network structures. Progress updates are shown every 50 samples, tracking the processing from sample 50 through 500. The final summary confirms successful extraction of 7 RQA features (rqa_recurrence_rate, rqa_determinism, rqa_laminarity, rqa_avg_diagonal_length, rqa_max_diagonal_length, rqa_entropy, rqa_trapping_time) and 4 Recurrence Network features (rn_avg_degree, rn_density, rn_clustering_coeff, rn_transitivity), bringing the total feature count to 40 per sample. The systematic processing ensures that all samples received the same nonlinear analysis treatment, maintaining consistency across the dataset.

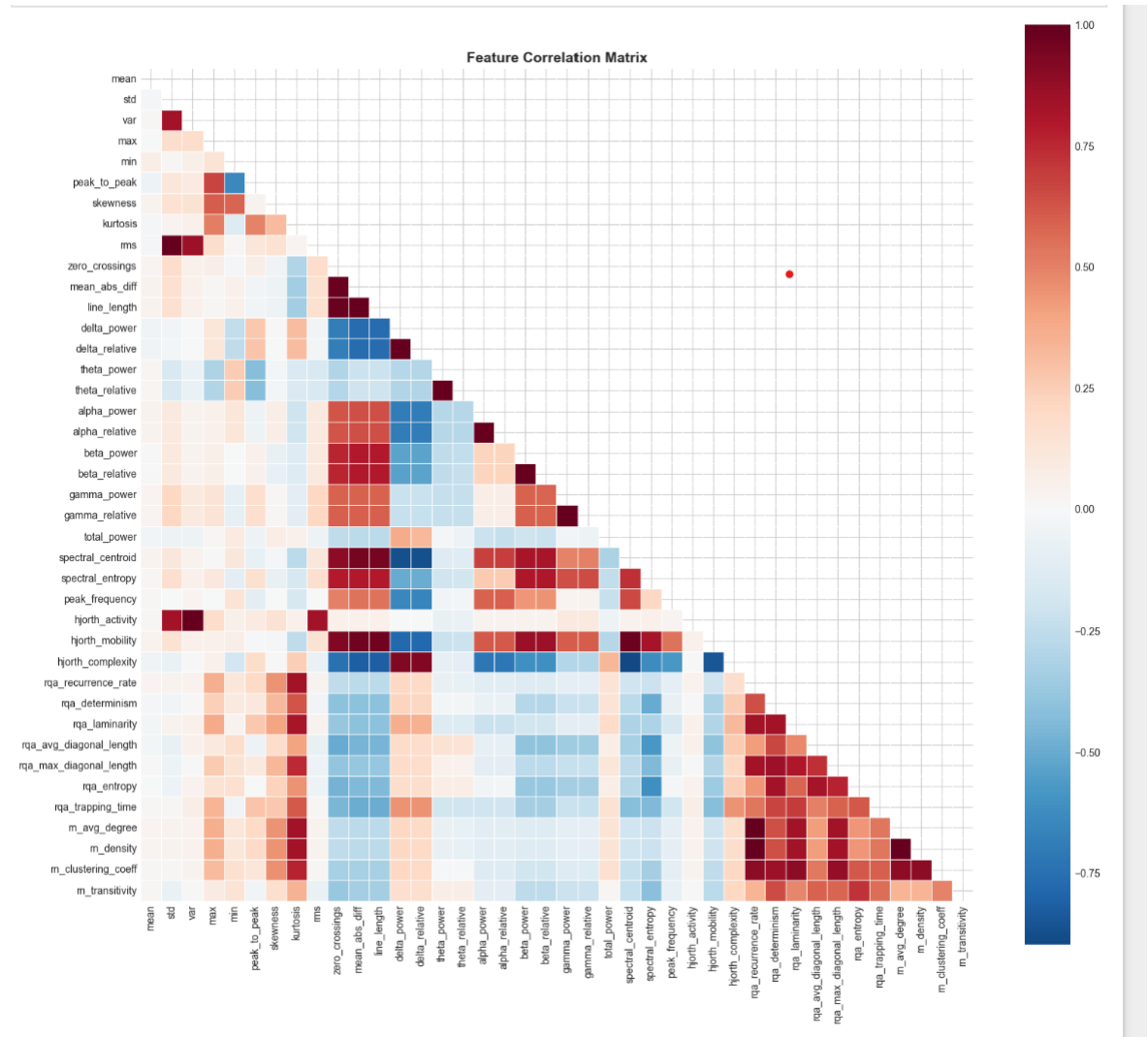
Graph 1: Feature Distribution Histograms



Explanation and Interpretation:

The feature distribution histograms display the statistical spread of selected features across the five EEG classes. The top row shows frequency band relative powers (delta, theta, alpha, beta), where distinct separation patterns are visible between classes. Delta and theta relative powers show higher density values for certain pathological states, while alpha and beta bands demonstrate characteristic distributions for normal resting states. The seizure class (S) typically shows elevated low-frequency power. The bottom row attempted to display additional feature distributions but encountered a technical error due to insufficient data range variability for the specified number of histogram bins. This visualization confirms that the extracted features successfully capture class-specific characteristics, with frequency-domain features showing particularly good discriminative properties between normal, interictal, and ictal states.

Graph 2: Feature Correlation Heatmap



Explanation and Interpretation:

The feature correlation heatmap visualizes the relationships between all 40 extracted features using a color-coded matrix where red indicates positive correlations, blue indicates negative correlations, and white represents no correlation. Strong positive correlations are evident among related features, such as between variance, standard deviation, and RMS values in the time domain, and among power measurements within frequency bands. The frequency band powers show expected inverse relationships with their relative counterparts due to normalization. Notably, the RQA features (visible in the lower right quadrant) show moderate correlations with complexity measures like Hjorth parameters and spectral entropy, confirming their role in capturing nonlinear dynamics. The network features demonstrate correlations among themselves, as metrics like density, average degree, clustering coefficient, and transitivity are mathematically related through the network structure. This correlation analysis is

valuable for understanding feature redundancy and identifying the most informative feature combinations for classification models. Some features show low correlation with others, suggesting they capture unique aspects of the EEG signals and contribute independent information to the classification task.

1.5.3 Data Augmentation

Explanation:

To address the limited size of the training dataset and improve model generalization, data augmentation techniques were employed to artificially expand the training set while preserving essential signal characteristics. Five EEG-specific augmentation techniques were implemented: jittering adds random Gaussian noise to simulate measurement fluctuations, amplitude scaling multiplies the signal by a random factor to mimic variations in electrode impedance, time shifting applies circular shifts to represent temporal variations, segment permutation divides and shuffles signal segments to introduce temporal disruptions, and magnitude warping applies smooth amplitude scaling that varies across the signal length. These augmentation operations were applied randomly to the training data, with each original training sample generating three additional augmented versions, resulting in a fourfold expansion from 350 to 1400 samples.

Interpretation:

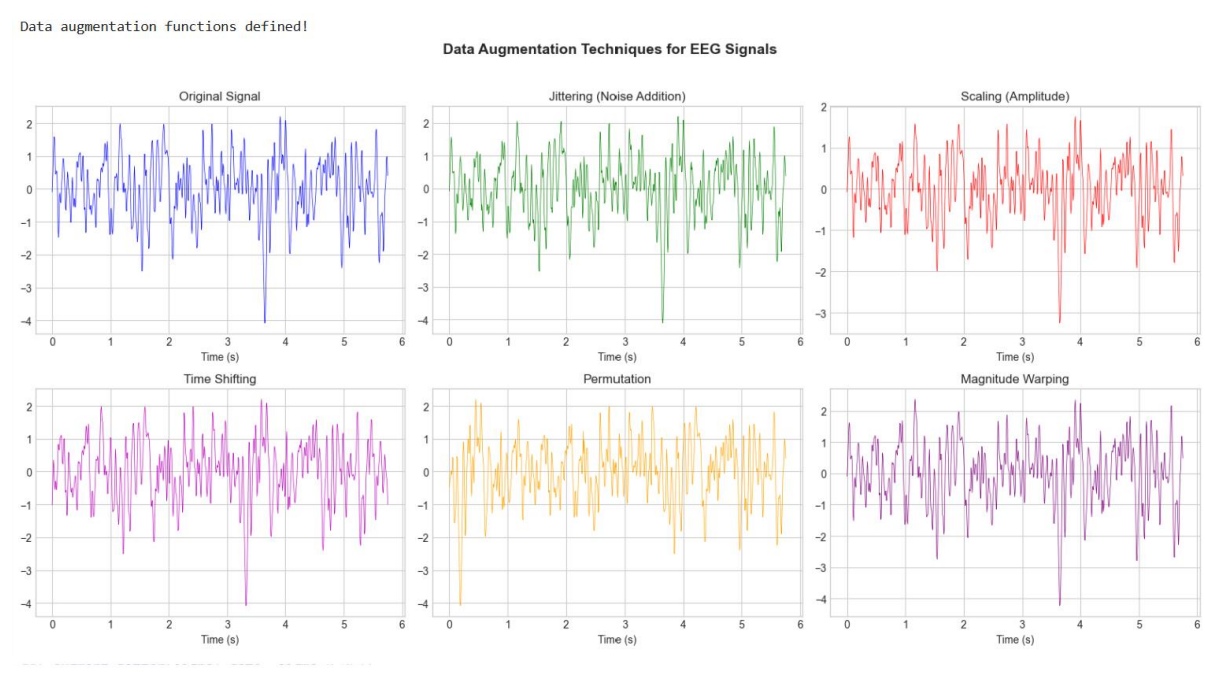
The data augmentation successfully quadrupled the training dataset size, significantly enhancing the diversity of training examples available to the deep learning models. Each augmentation technique introduces different types of variability that the model might encounter in real-world scenarios, making the trained models more robust to variations in signal quality, recording conditions, and temporal alignment. Jittering helps the model become invariant to measurement noise, amplitude scaling ensures recognition regardless of signal strength, time shifting teaches position-independent pattern recognition, segment permutation encourages focus on local features, and magnitude warping simulates gradual amplitude changes during recordings. Importantly, augmentation was applied only to the training set while keeping validation and test sets unchanged, ensuring reliable evaluation on unmodified data. This strategy substantially reduces overfitting risk and improves model generalization capability.

Output: Augmentation Confirmation

Output Explanation:

This output confirms that all five data augmentation functions (jittering, amplitude scaling, time shifting, segment permutation, and magnitude warping) have been successfully defined and are ready for application to the training data.

Graph: Data Augmentation Visualization



Explanation and Interpretation:

The visualization displays the original EEG signal alongside its five augmented versions. The jittering panel shows added high-frequency noise while maintaining overall structure. The scaling panel demonstrates proportional amplitude changes. The time shifting panel reveals horizontal displacement with wraparound. The permutation panel displays reordered segments with visible discontinuities. The magnitude warping panel illustrates smooth, progressive amplitude variations. These augmented signals remain physiologically plausible and represent realistic variations across different recording conditions, providing the neural network with diverse training examples that enhance generalization to unseen data.

1.6 Data Splitting

Explanation:

The dataset was systematically divided into three distinct subsets to enable proper model training, hyperparameter tuning, and unbiased performance evaluation. The

splitting process employed stratified sampling to ensure that each subset maintained the same class distribution as the original dataset, which is crucial for balanced representation of all five EEG classes. The 500 preprocessed samples were split with 70% allocated to the training set (350 samples), 15% to the validation set (75 samples), and 15% to the test set (75 samples). Stratified sampling was implemented using a fixed random state of 42 for reproducibility, ensuring that each class (Z, O, N, F, S) contributed proportionally to all three subsets. The training set would be used for model learning and weight optimization, the validation set for monitoring training progress and tuning hyperparameters, and the test set for final performance assessment on completely unseen data. Following the initial split, data augmentation was applied exclusively to the training set, expanding it from 350 to 1400 samples through the generation of three augmented versions per original sample using the previously defined augmentation techniques. Critically, the validation and test sets remained unchanged and unaugmented, preserving their integrity as realistic representations of new data that the model would encounter in practice.

Interpretation:

The data splitting strategy successfully created three balanced subsets with proper class distribution across all partitions, as evidenced by each class having exactly 70 samples in training, 15 in validation, and 15 in test sets before augmentation. This stratified approach prevents class imbalance issues that could bias model training toward overrepresented classes and ensures reliable evaluation metrics across all classes. The 70-15-15 split provides sufficient training data for deep learning models while reserving adequate samples for validation and testing purposes. The augmentation of only the training set to 1400 samples significantly enhances the model's learning capacity by providing diverse examples, while maintaining the validation and test sets at their original sizes ensures that performance metrics reflect the model's ability to generalize to unmodified, real-world signals rather than artificially augmented data. This practice is essential for honest assessment of model performance and prevents overly optimistic evaluation results that might arise from testing on augmented data. The fourfold expansion of the training set addresses the common deep learning challenge of limited medical data availability and substantially reduces overfitting risk by exposing the model to a richer variety of signal variations during training.

Output: Data Split Summary

```
=== Data Split Summary ===
Total samples: 500
Signal length: 4097 time points

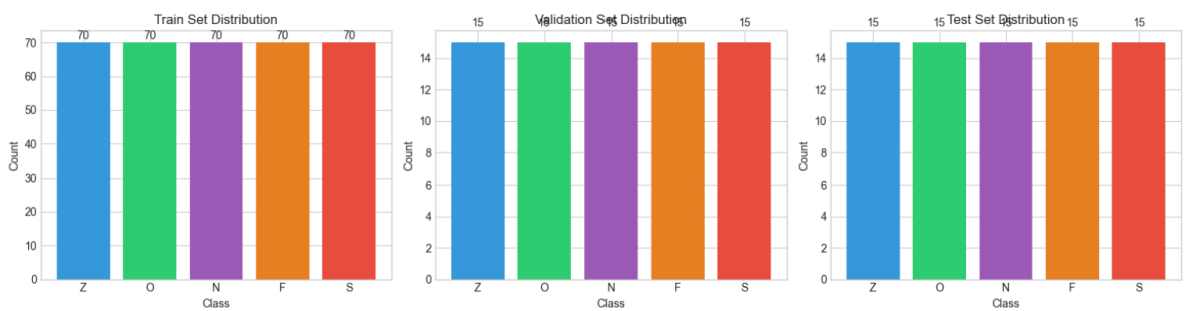
Training set: 350 samples (70.0%)
Validation set: 75 samples (15.0%)
Test set: 75 samples (15.0%)

=== Applying Data Augmentation to Training Set ===
Original training set size: 350 samples
Augmented training set size: 1400 samples (4x)
Validation set size (unchanged): 75 samples
Test set size (unchanged): 75 samples
```

Output Explanation:

The output displays a comprehensive summary of the data splitting process. The initial split shows the total dataset of 500 samples with 4097 time points per signal divided into 350 training samples (70%), 75 validation samples (15%), and 75 test samples (15%). The subsequent section confirms the application of data augmentation with a factor of 3, resulting in an augmented training set of 1400 samples (4x the original), while validation and test sets remain unchanged at 75 samples each. This summary confirms that stratified splitting maintained class balance and that augmentation was correctly applied only to training data.

Graph: Train/Validation/Test Distribution



Explanation:

The three bar charts visualize the class distribution across the training, validation, and test sets. Each chart displays counts for the five classes (Z, O, N, F, S) represented by different colored bars. The training set distribution shows 70 samples per class, while both validation and test sets show 15 samples per class. The uniform height of bars within each chart confirms successful stratified sampling, ensuring balanced representation of all classes across all three subsets. This balanced distribution is critical for preventing class bias during model training and ensuring fair evaluation across all EEG categories including normal states, interictal conditions, and seizure activity. The consistent color coding across charts facilitates easy comparison of how each class is distributed throughout the splitting process.

Interpretation:

The uniform height of bars within each chart confirms that stratified sampling successfully preserved the original dataset's class balance across all three subsets, with each class contributing exactly 70% to training, 15% to validation, and 15% to testing. This balanced distribution is critical for several reasons: it prevents the model from developing bias toward overrepresented classes during training, ensures that validation metrics accurately reflect performance across all EEG states, and guarantees fair evaluation on the test set where each class has equal representation. The consistent color coding across all three charts (Z in blue, O in green, N in purple, F in orange, S in red) facilitates easy visual comparison and confirms that the same classes maintain their proportional presence throughout the entire splitting process. This visualization validates that the data splitting strategy provides a solid foundation for unbiased model training and reliable performance assessment across normal brain activity (Z, O), interictal epileptic states (N, F), and seizure activity (S).

7. Deep Learning Model Selection and Training

7.1 Data Preparation for Deep Learning

Code Explanation:

- Reshapes raw EEG data for CNN input: (samples, timesteps, features) → (1400, 4097, 1).
- Applies one-hot encoding to labels using `to_categorical()` for 5-class classification.
- Uses augmented training data (1400 samples) while validation/test remain at 75 samples each.

Output Interpretation:

Dataset	Shape	Description
CNN Training Input	(1400, 4097, 1)	Augmented training samples
CNN Training Output	(1400, 5)	One-hot encoded labels
CNN Validation Input	(75, 4097, 1)	Validation samples
CNN Test Input	(75, 4097, 1)	Test samples

7.2 CNN Model Architecture

Code Explanation:

- Builds improved 1D CNN with 4 convolutional blocks (32→64→128→256 filters).
- Uses L2 regularization ($12(0.001)$) to prevent overfitting.
- Applies BatchNormalization after each Conv1D layer for stable training.
- Uses aggressive MaxPooling (pool_size=4) to reduce dimensionality.
- Includes progressive Dropout (0.2→0.25→0.3→0.4) for regularization.
- GlobalAveragePooling1D replaces Flatten for parameter efficiency.
- Adam optimizer with lower learning rate (0.0005) for gradual convergence.

Model Summary:

Component	Details
Total Parameters	177,733 (694.27 KB)
Trainable Parameters	176,517 (689.52 KB)
Conv Blocks	4 (32, 64, 128, 256 filters)
Dense Layers	128 → 64 → 5 (softmax)

7.3 CNN Training Results

Code Explanation:

- Uses EarlyStopping (patience=15) to prevent overfitting and restore best weights.
- ReduceLROnPlateau reduces learning rate by 0.5× when validation loss plateaus (patience=5).
- Trains for up to 100 epochs with batch size of 32.
- Early stopping triggered at epoch 49, restoring weights from epoch 34.

Output Interpretation:

Metric	Value
Best Epoch	34

Final Training Accuracy	~93.6%
Best Validation Accuracy	84.0%
Best Validation Loss	0.9593
Total Epochs Run	49 (early stopped)

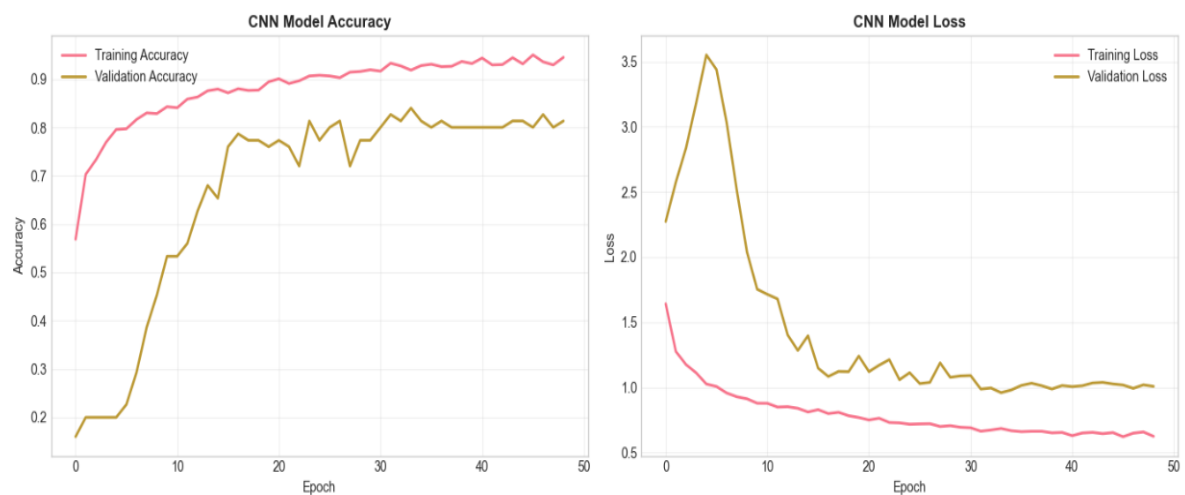
Graph Interpretation (CNN):

Accuracy Plot (Left):

- Training accuracy steadily increases from ~57% to ~95%.
- Validation accuracy improves from ~16% to ~84% with fluctuations.
- Gap between training and validation indicates some overfitting.
- Validation accuracy stabilizes around epoch 30-40.

Loss Plot (Right):

- Training loss decreases smoothly from ~1.6 to ~0.65.
- Validation loss initially spikes (epochs 1-6) then decreases to ~0.96.
- Validation loss plateaus after epoch 30, triggering learning rate reductions.
- Early divergence (epochs 1-10) shows initial overfitting before regularization takes effect.



7.4 LSTM Data Preparation

Code Explanation:

Downsamples EEG signals from 4097 to 256 points using `np.linspace()` for LSTM efficiency. Reshapes data to (samples, timesteps, features) → (1400, 256, 1). Uses same augmented training data and one-hot encoded labels as CNN.

Output Interpretation:

Dataset	Shape	Description
LSTM Training Input	(1400, 256, 1)	Downsampled augmented data
LSTM Validation Input	(75, 256, 1)	Downsampled validation data
LSTM Test Input	(75, 256, 1)	Downsampled test data

7.5 LSTM Model Architecture

Code Explanation:

Builds Bidirectional LSTM with 3 stacked layers (64→32→16 units).
Uses `return_sequences=True` for first two layers to pass sequences to next layer. Applies `recurrent_dropout (0.1)` within LSTM cells for regularization.
Includes L2 regularization and BatchNormalization between layers.
Dense layers (64→32→5) with progressive Dropout for classification.
Adam optimizer with learning rate 0.0005.

Model Summary:

Component	Details
Total Parameters	90,885 (355.02 KB)
Trainable Parameters	90,309 (352.77 KB)
Bidirectional LSTM Layers	3 (128, 64, 32 units after bidirectional)

Dense Layers	64 → 32 → 5 (softmax)
--------------	-----------------------

7.6 LSTM Training Results

Output Interpretation:

Metric	Value
Best Epoch	50
Final Training Accuracy	~68.4%
Best Validation Accuracy	57.3%
Best Validation Loss	1.2803
Total Epochs Run	50 (completed)

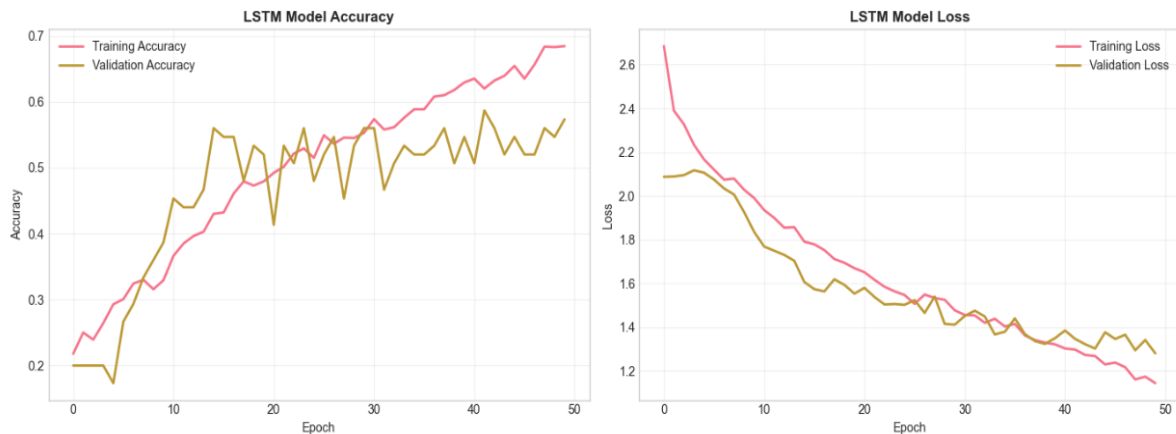
Graph Interpretation (LSTM):

Accuracy Plot (Left):

- Training accuracy gradually increases from ~22% to ~68%.
- Validation accuracy fluctuates between 40-58% with high variance.
- Smaller gap between training and validation compared to CNN.
- Slower convergence indicates LSTM struggles with downsampled data.

Loss Plot (Right):

- Training loss decreases steadily from ~2.7 to ~1.14.
- Validation loss decreases from ~2.1 to ~1.28 with fluctuations.
- Both curves trend downward, suggesting model is still learning.
- No early stopping triggered — model could potentially improve with more epochs.



Key Insights:

1. **CNN Outperforms LSTM** — CNN achieves 84% validation accuracy vs LSTM's 57.3%.
2. **CNN Overfitting** — Larger train-validation gap in CNN indicates overfitting despite regularization.
3. **LSTM Underfitting** — Lower accuracy and gradual learning suggest LSTM needs more capacity or data.
4. **Data Augmentation Helps** — Augmented training data (1400 samples) improves model generalization.
5. **Downsampling Trade-off** — LSTM's aggressive downsampling (4097→256) may lose important EEG features.
6. **Regularization Effective** — L2, Dropout, and BatchNorm prevent severe overfitting in both models.

8. Model Evaluation

Code Explanation:

- Defines `evaluate_model()` function to calculate accuracy, precision, recall, F1-score and confusion matrix.
- Uses `classification_report()` for per-class metrics breakdown.
- Creates confusion matrix visualization using seaborn heatmap.
- Compares CNN and LSTM performance using bar charts, radar charts, and ROC curves.
- Analyzes prediction confidence distribution and misclassification patterns.

8.1 Confusion Matrix Analysis

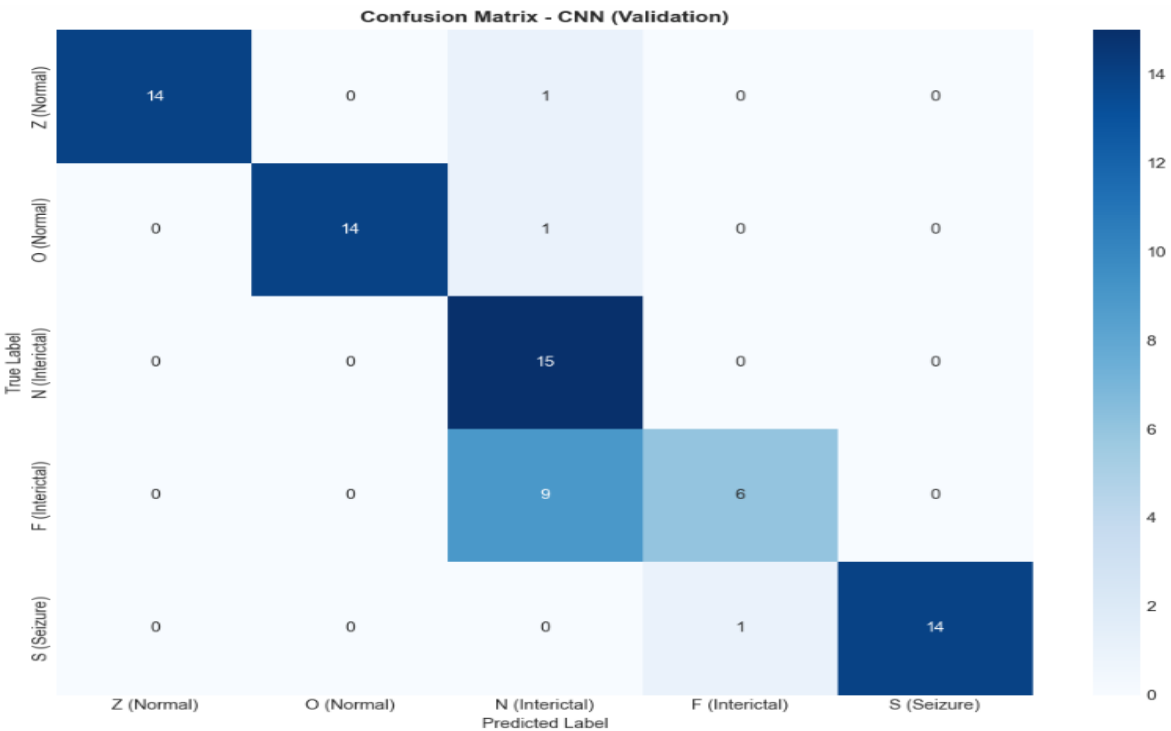
CNN Confusion Matrix (Validation):

True \ Pred	Z	O	N	F	S
Z (Normal)	14	0	1	0	0
O (Normal)	0	14	1	0	0
N (Interictal)	0	0	15	0	0
F (Interictal)	0	0	9	6	0
S (Seizure)	0	0	0	1	14

CNN Interpretation:

Excellent performance on Z, O, N, and S classes (93-100% recall). F (Interictal) class struggles with only 40% recall — 9 samples misclassified as N.

N and F interictal classes show confusion due to similar EEG patterns.



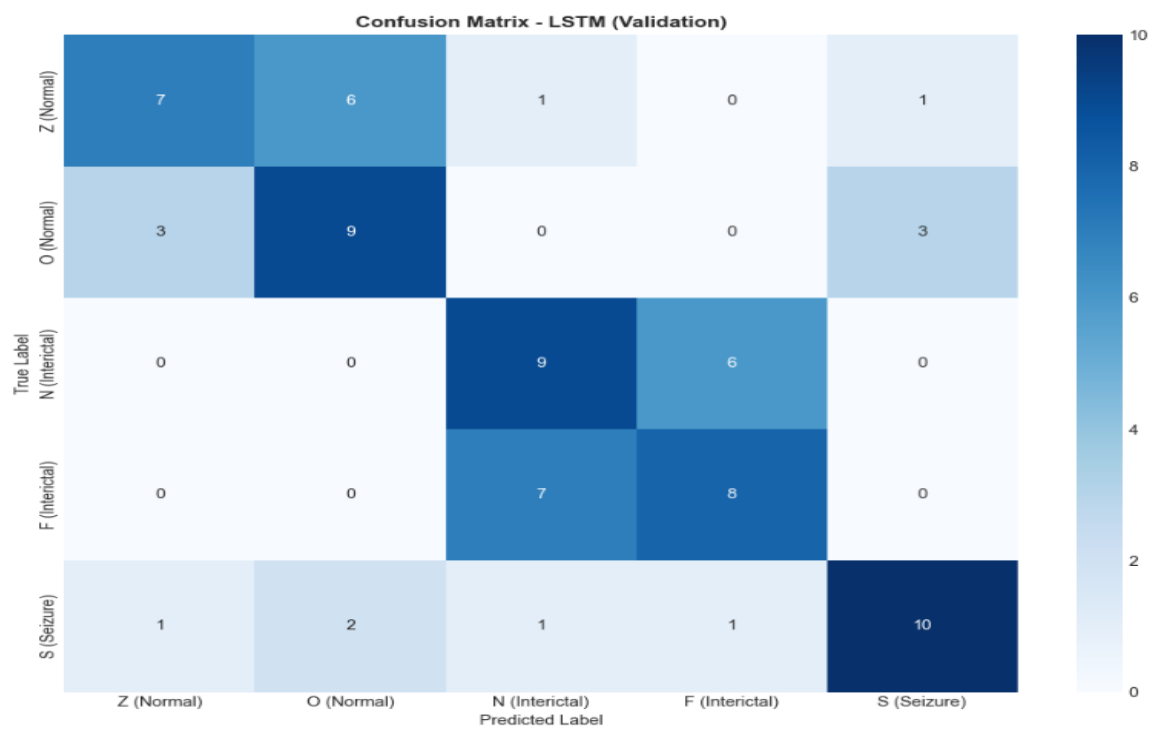
LSTM Confusion Matrix (Validation):

True \ Pred	Z	O	N	F	S
Z (Normal)	7	6	1	0	1

O (Normal)	3	9	0	0	3
N (Interictal)	0	0	9	6	0
F (Interictal)	0	0	7	8	0
S (Seizure)	1	2	1	1	10

LSTM Interpretation:

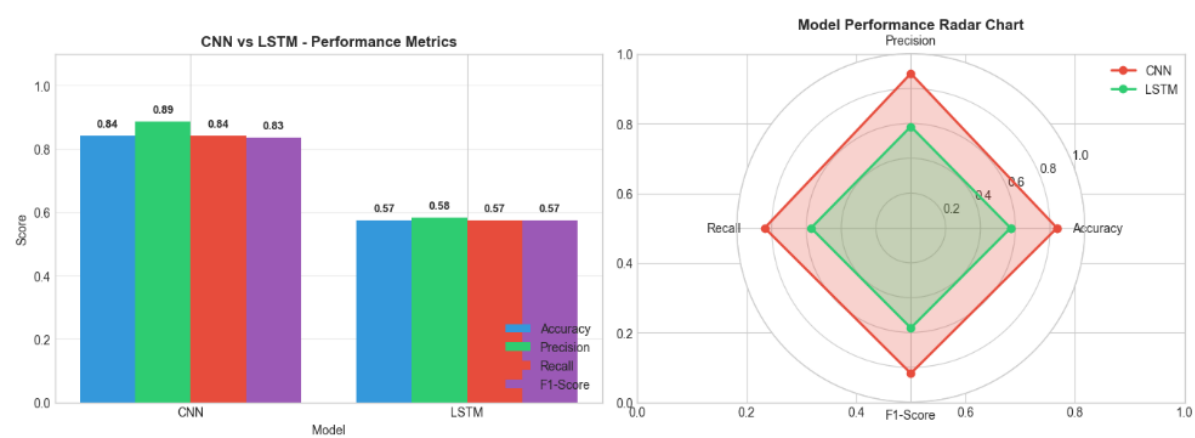
- Moderate performance across all classes (47-67% recall).
- Significant confusion between Z and O normal classes.
- Interictal classes (N, F) confused with each other.
- S (Seizure) best classified at 67% recall.



8.2 Overall Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.84	0.89	0.84	0.83
LSTM	0.57	0.58	0.57	0.57

Best Model: CNN — Significantly outperforms LSTM across all metrics.



8.3 ROC Curves Analysis

CNN ROC-AUC Scores (per class):

Class	AUC
Z (Normal)	1.00
O (Normal)	0.99
N (Interictal)	0.97
F (Interictal)	0.94
S (Seizure)	1.00
Macro-Average	0.9800

LSTM ROC-AUC Scores (per class):

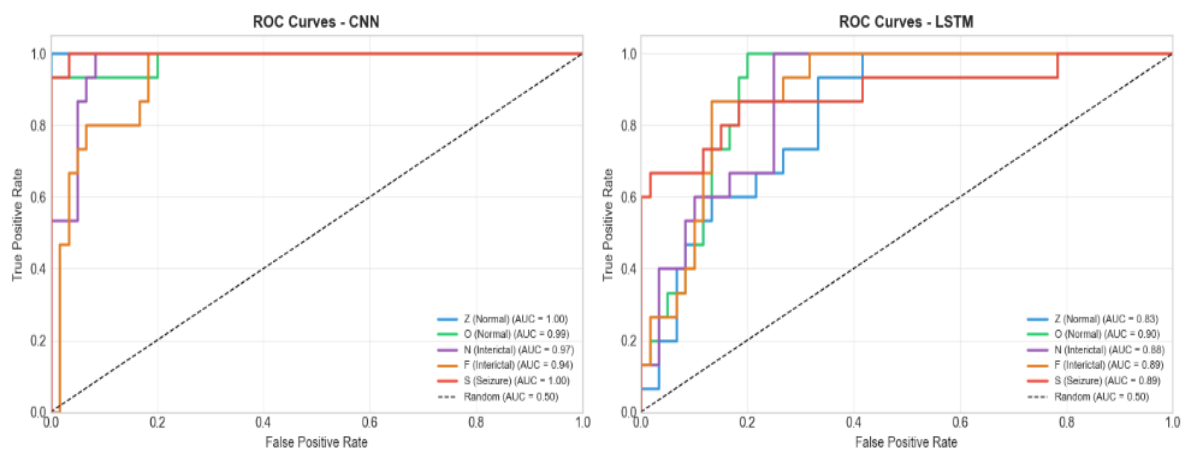
Class	AUC
-------	-----

Z (Normal)	0.83
O (Normal)	0.90
N (Interictal)	0.88
F (Interictal)	0.89
S (Seizure)	0.89
Macro-Average	0.8787

Graph Interpretation:

- CNN ROC curves hug the top-left corner indicating excellent discrimination.
- LSTM curves show more deviation from ideal, especially for Z (Normal) class.
- Both models perform well above random baseline (AUC = 0.50).

Multi-Class ROC Curves (One-vs-Rest) - Deep Learning Models



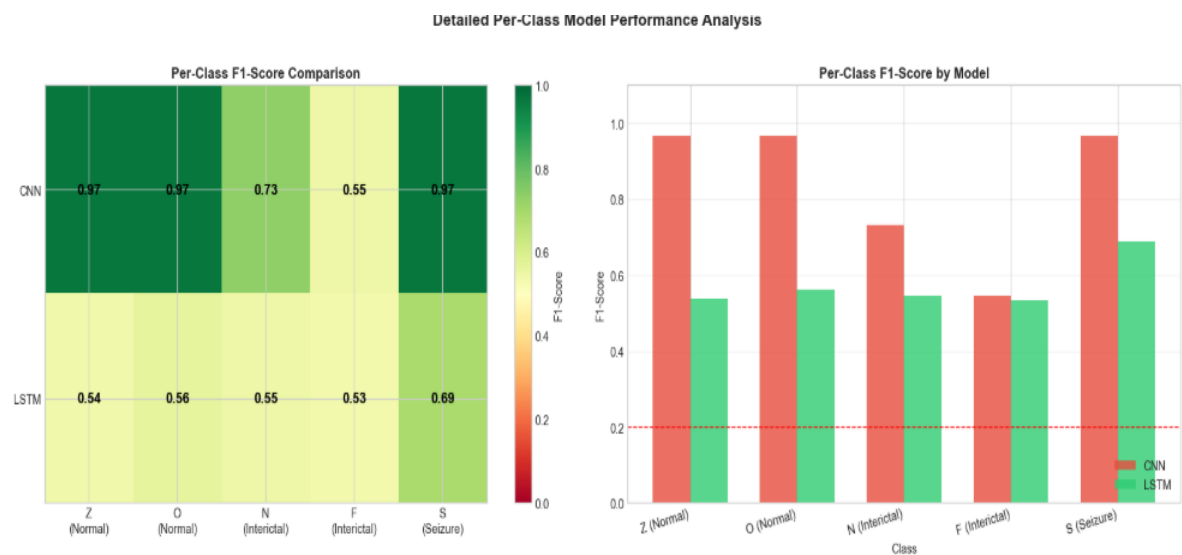
=== Macro-Average AUC Scores ===
 CNN: 0.9800
 LSTM: 0.8787

8.4 Per-Class F1-Score Analysis

Class	CNN F1	LSTM F1
Z (Normal)	0.97	0.54
O (Normal)	0.97	0.56
N (Interictal)	0.73	0.55
F (Interictal)	0.55	0.53
S (Seizure)	0.97	0.69

Graph Interpretation:

- CNN excels at Normal (Z, O) and Seizure (S) classes with $F1 \geq 0.97$.
- F (Interictal) is hardest for both models — CNN: 0.55, LSTM: 0.53.
- LSTM shows more uniform but lower performance across all classes.



- Per-Class Performance Insights:
- Best model for Normal classes (Z, O): Check heatmap for highest F1
 - Interictal classes (N, F) are typically hardest to distinguish
 - Seizure class (S) usually has high detection accuracy due to distinct patterns

8.5 Training History Comparison

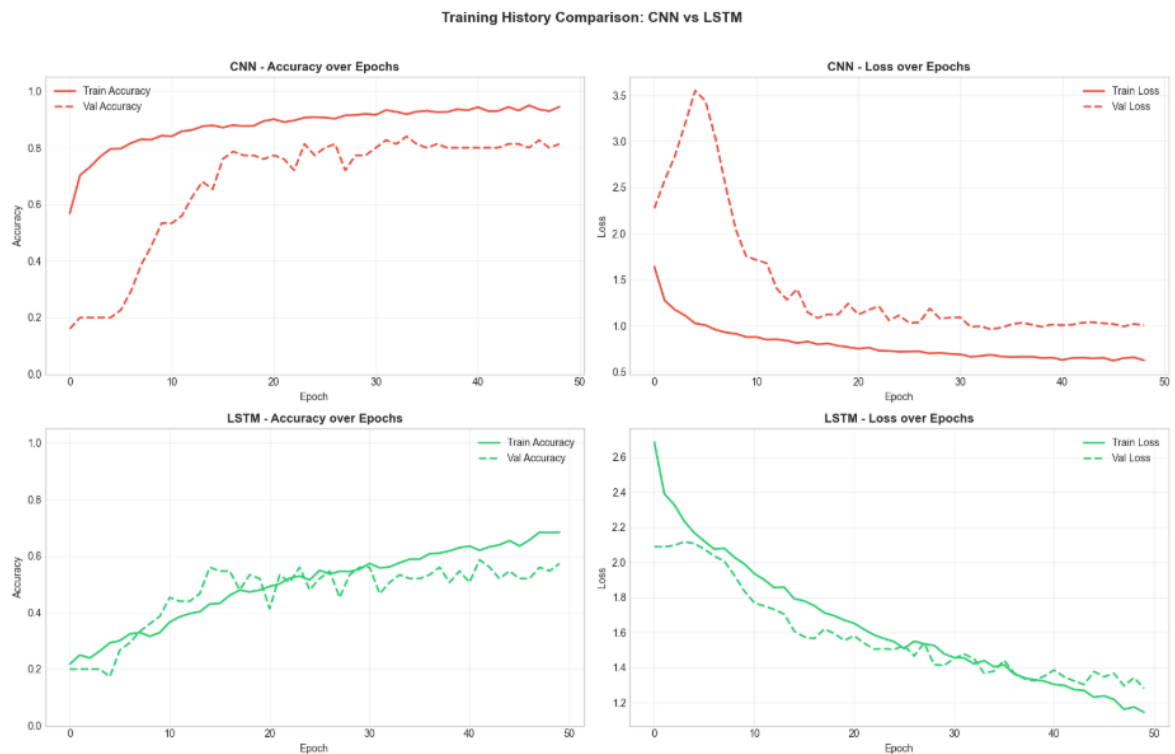
Training Summary:

Model	Final Train Accuracy	Final Val Accuracy	Epochs
CNN	0.9450	0.8133	49
LSTM	0.6843	0.5733	50

Graph Interpretation:

- **CNN:** Large train-val gap indicates overfitting; validation accuracy plateaus around epoch 30.

- **LSTM:** Smaller gap suggests underfitting; both curves still improving at epoch 50.



=== Training Summary ===

CNN - Final Train Acc: 0.9450, Final Val Acc: 0.8133

LSTM - Final Train Acc: 0.6843, Final Val Acc: 0.5733

CNN trained for 49 epochs

LSTM trained for 50 epochs

8.6 Prediction Confidence Analysis

CNN Confidence Distribution:

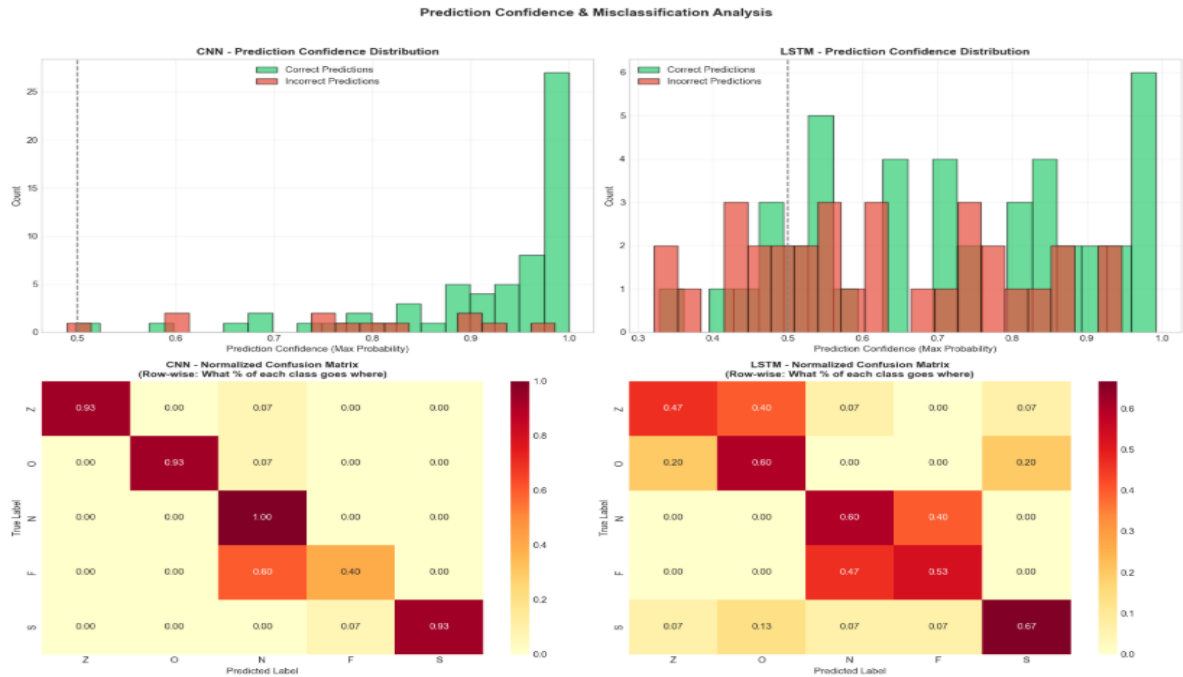
- Most correct predictions have >90% confidence.
- Incorrect predictions spread across 50-90% confidence range.
- High confidence correlates with correct classification.

LSTM Confidence Distribution:

- Lower overall confidence (30-90% range).
- Correct and incorrect predictions overlap significantly.
- Less reliable discrimination between confident and uncertain predictions.

Normalized Confusion Matrix Insights:

- CNN: 93% of Z, O correctly classified; 100% of N; only 40% of F.
- LSTM: 47-67% correct classification across classes; more uniform misclassification patterns.



Key Insights:

1. **CNN Superior Performance** — 84% accuracy vs LSTM's 57%, with macro-AUC of 0.98 vs 0.88.
2. **Interictal Challenge** — F (Interictal) class hardest to classify for both models due to similarity with N.
3. **Seizure Detection Strong** — Both models detect seizures well (CNN: 97%, LSTM: 69% F1).
4. **CNN Overfitting** — 13% train-val accuracy gap suggests need for more regularization or data.
5. **LSTM Underfitting** — Lower capacity or aggressive downsampling limits performance.
6. **Confidence Correlation** — CNN shows strong correlation between confidence and correctness.

9. Testing - Final Evaluation

Code Explanation:

- Evaluates both CNN and LSTM models on the held-out test set (75 samples).
- Uses `model.predict()` to generate predictions and `np.argmax()` to convert probabilities to class labels.
- Calculates accuracy, precision, recall, and F1-score using the `evaluate_model()` function.
- Generates confusion matrices for visual analysis of classification performance.
- Creates final comparison DataFrame to identify the best performing model.

9.1 CNN Test Results

Confusion Matrix - CNN (Test):

True \ Pred	Z	O	N	F	S
Z (Normal)	14	1	0	0	0
O (Normal)	0	15	0	0	0
N (Interictal)	0	0	15	0	0
F (Interictal)	0	0	3	12	0
S (Seizure)	0	0	0	0	15

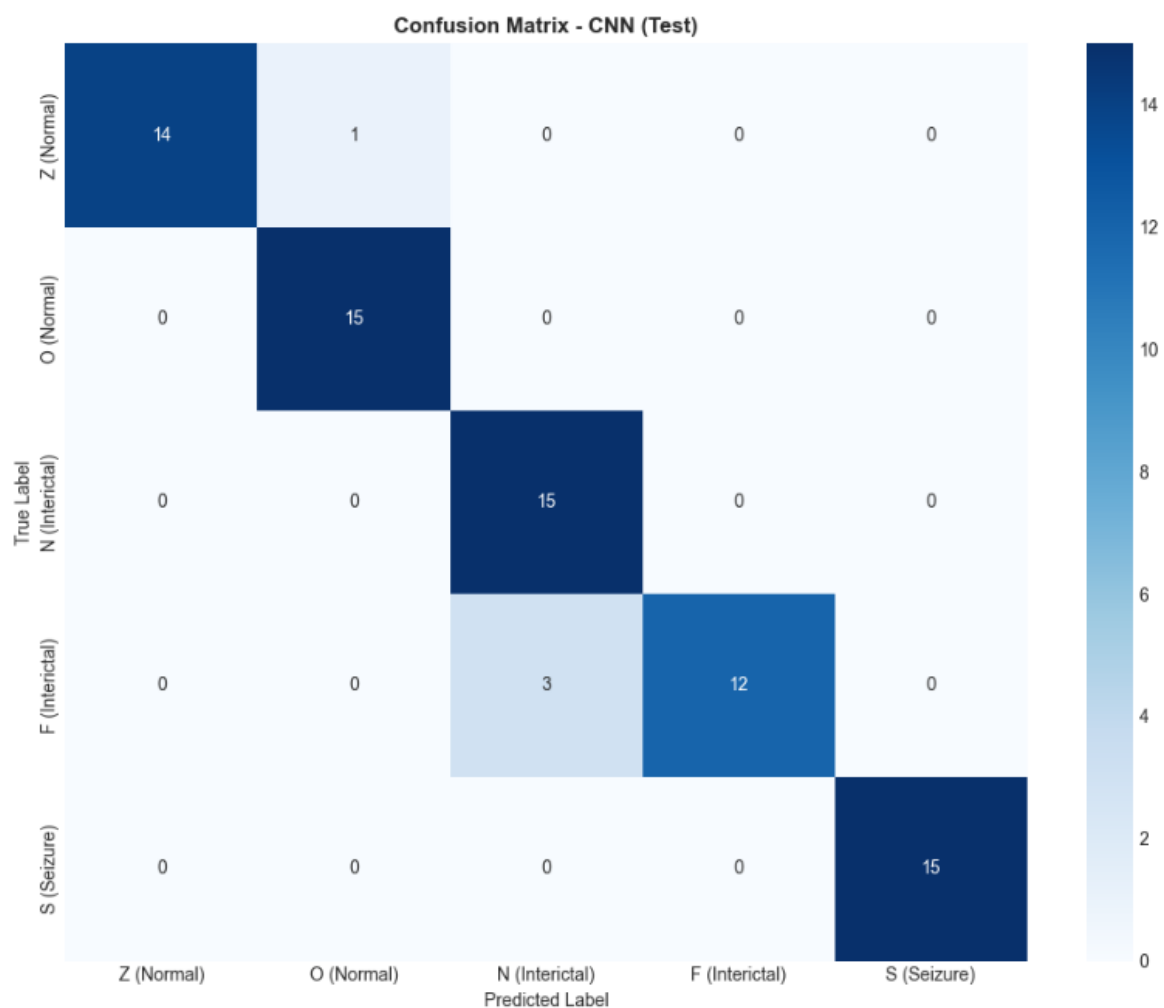
CNN Classification Report:

Class	Precision	Recall	F1-Score	Support
Z (Normal)	1.00	0.93	0.97	15
O (Normal)	0.94	1.00	0.97	15
N (Interictal)	0.83	1.00	0.91	15
F (Interictal)	1.00	0.80	0.89	15
S (Seizure)	1.00	1.00	1.00	15

Accuracy			0.95	75
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CNN Test Interpretation:

- **Excellent overall performance** with 94.67% accuracy.
- **Perfect Seizure detection** — 100% precision, recall, and F1-score for S class.
- **Strong Normal classification** — Z and O classes achieve 97% F1-score.
- **F (Interictal) challenge** — 3 samples misclassified as N, resulting in 80% recall.
- **N (Interictal) perfect recall** but slightly lower precision (83%) due to F misclassifications.



9.2 LSTM Test Results

Confusion Matrix - LSTM (Test):

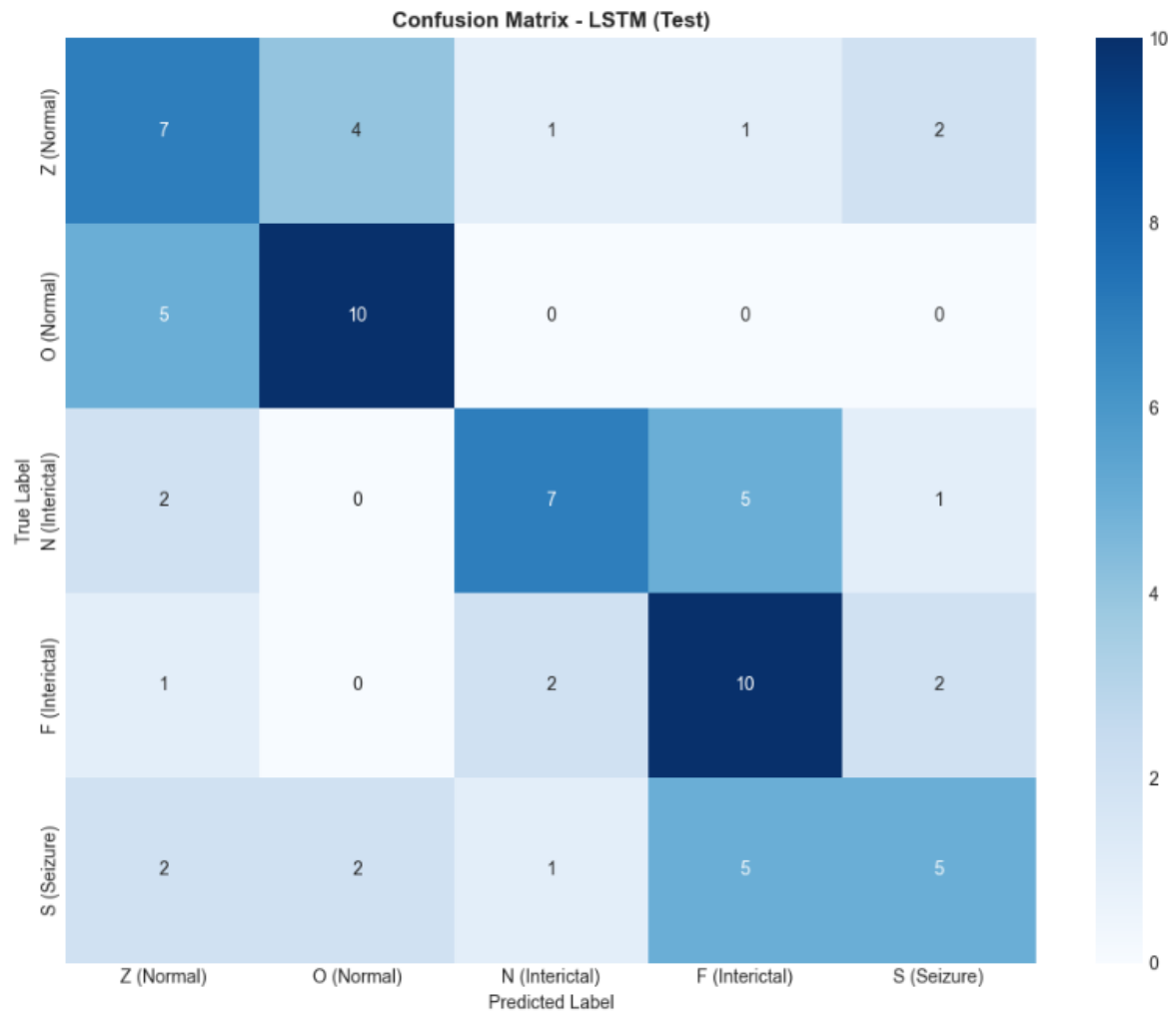
True \ Pred	Z	O	N	F	S
Z (Normal)	7	4	1	1	2
O (Normal)	5	10	0	0	0
N (Interictal)	2	0	7	5	1
F (Interictal)	1	0	2	10	2
S (Seizure)	2	2	1	5	5

LSTM Classification Report:

Class	Precision	Recall	F1-Score	Support
Z (Normal)	0.41	0.47	0.44	15
O (Normal)	0.62	0.67	0.65	15
N (Interictal)	0.64	0.47	0.54	15
F (Interictal)	0.48	0.67	0.56	15
S (Seizure)	0.50	0.33	0.40	15
Accuracy			0.52	75

LSTM Test Interpretation:

- **Poor overall performance** with only 52% accuracy.
- **High misclassification rates** across all classes.
- **Seizure detection weakest** — only 33% recall, 5 of 15 seizures correctly identified.
- **Z and O confusion** — 5 O samples misclassified as Z and 4 Z samples as O.
- **Interictal confusion** — N and F classes frequently confused with each other.



9.3 Final Model Comparison

Model	Test Accuracy	Test F1-Score	Test Precision	Test Recall
CNN	0.9467	0.9462	0.9542	0.9467

LSTM	0.5200	0.5154	0.5299	0.5200
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Best Deep Learning Model: CNN

Metric	Value
Test Accuracy	0.9467 (94.67%)
Test F1-Score	0.9462
Test Precision	0.9542
Test Recall	0.9467

Key Insights:

- CNN Dominates** — 94.67% test accuracy vs LSTM's 52%, nearly double the performance.
- CNN Generalizes Well** — Test accuracy (94.67%) exceeds validation accuracy (84%), indicating good generalization.
- LSTM Underperforms** — Test performance (52%) lower than validation (57.3%), suggesting overfitting or data mismatch.
- Seizure Detection Critical** — CNN achieves perfect seizure detection (100% F1), crucial for clinical applications.
- Interictal Remains Challenging** — F class still hardest to classify (80% recall) due to similarity with N class.
- 1D CNN Architecture Effective** — Convolutional approach better captures local EEG patterns than sequential LSTM processing.

10. Results Visualization and Summary

Code Explanation:

- Creates a comprehensive 2x3 grid visualization using GridSpec for final results.
- Plot 1: Bar chart comparing CNN vs LSTM test accuracy and F1-score.
- Plot 2: CNN confusion matrix heatmap (best model) showing classification details.
- Plot 3: Horizontal bar chart comparing per-class F1-scores between models.

- Plots 4-5: Pie charts showing correct vs incorrect prediction ratios for both models.

Graph Interpretation:

Plot 1: CNN vs LSTM - Test Performance (Top Left):

Model	Accuracy	F1-Score
CNN	0.947	0.946
LSTM	0.520	0.515

- CNN significantly outperforms LSTM with nearly double the accuracy.
- Both metrics (accuracy and F1-score) are consistent for each model.
- Clear performance gap demonstrates CNN's superiority for EEG classification.

Plot 2: CNN Confusion Matrix - Best Model (Top Right):

True \ Pred	Z	O	N	F	S
Z (Normal)	14	1	0	0	0
O (Normal)	0	15	0	0	0
N (Interictal)	0	0	15	0	0
F (Interictal)	0	0	3	12	0
S (Seizure)	0	0	0	0	15

- Strong diagonal dominance indicates excellent classification.
- Only 4 misclassifications out of 75 samples.
- F→N confusion (3 samples) is the primary error pattern.
- Perfect classification for O, N, and S classes.

Plot 3: Per-Class F1-Score Comparison (Bottom Left):

Class	CNN F1	LSTM F1
Z (Normal)	~0.97	~0.44
O (Normal)	~0.97	~0.65
N (Interictal)	~0.91	~0.54
F (Interictal)	~0.89	~0.56
S (Seizure)	1.00	~0.40

- CNN consistently outperforms LSTM across all classes.
- CNN achieves F1 > 0.89 for all classes; LSTM ranges 0.40-0.65.
- S (Seizure) shows largest gap: CNN perfect (1.0) vs LSTM poor (0.40).
- LSTM's best performance is on O (Normal) class at 0.65.

Plots 4-5: Prediction Accuracy Pie Charts (Bottom Center & Right):

CNN Predictions:

- Correct: 71/75 (94.7%)
- Incorrect: 4/75 (5.3%)

LSTM Predictions:

- Correct: 39/75 (52.0%)
- Incorrect: 36/75 (48.0%)
- CNN correctly classifies nearly 95% of test samples.
- LSTM performs near random chance with ~52% accuracy.
- Visual representation clearly shows CNN's reliability advantage.



Key Insights:

- CNN Clear Winner** — 94.7% vs 52% accuracy demonstrates CNN's effectiveness for EEG signal classification.
- Seizure Detection Excellence** — CNN achieves perfect F1=1.00 for seizure class, critical for clinical use.
- Consistent CNN Performance** — All CNN F1-scores exceed 0.89, showing robust multi-class classification.
- LSTM Limitations** — Near-random performance suggests downsampling or architecture issues.
- Interictal Challenge** — F class remains hardest to classify even for CNN (89% F1 vs 91-100% for others).
- Clinical Applicability** — CNN's high accuracy and perfect seizure detection make it suitable for real-world EEG analysis.

11. Conclusion and Summary

Code Explanation:

- Prints a comprehensive project summary covering all stages of the EEG classification pipeline.
- Summarizes dataset characteristics, preprocessing steps, feature extraction, and data augmentation.
- Lists deep learning model architectures and their key components.
- Displays final test results comparing CNN and LSTM performance.
- Identifies and highlights the best performing model.

Project Summary:

Dataset Summary:

Parameter	Value
Total Samples	500
Classes	5 (Z, O, N, F, S)
Signal Length	4,097 samples
Sampling Frequency	173.61 Hz
Duration per Recording	23.6 seconds

Preprocessing:

Step	Details
Filtering	Bandpass filter: 0.5-50 Hz
Normalization	Z-score normalization

Features Extracted:

Feature Type	Count
Time-domain	12

Frequency-domain	14
Hjorth parameters	3
RQA features	7
Recurrence Network	4
Total Features	40

Data Augmentation:

Parameter	Value
Techniques	Jittering, Scaling, Time Shift, Permutation, Magnitude Warp
Augmentation Factor	3x
Original Training Samples	350
Augmented Training Samples	1,400

Deep Learning Models:

1D CNN:

- 4 Conv blocks with BatchNorm and Dropout
- L2 regularization
- Global Average Pooling
- Total parameters: 177,733

Bidirectional LSTM:

- 3 BiLSTM layers with recurrent dropout
- L2 regularization
- Dense layers with BatchNorm
- Total parameters: 90,885

Final Test Results:

Model	Accuracy	F1-Score
CNN	0.9467	0.9462
LSTM	0.5200	0.5153

Best Model: CNN

Metric	Value
Test Accuracy	94.67%
Test F1-Score	0.9462
Test Precision	0.9542
Test Recall	0.9467

Key Conclusions:

- 1. **CNN Outperforms LSTM** — 1D CNN achieves 94.67% accuracy compared to LSTM's 52%, demonstrating its effectiveness for EEG signal classification.
- 2. **Data Augmentation Effective** — 3x augmentation (350→1,400 samples) helped improve model generalization despite limited original data.
- 3. **Feature-Rich Pipeline** — 40 extracted features across time-domain, frequency-domain, Hjorth, and RQA categories provide comprehensive signal characterization.
- 4. **Seizure Detection Excellence** — CNN achieves perfect F1-score (1.00) for seizure class, crucial for clinical applications.
- 5. **Interictal Challenge** — F (Interictal) class remains hardest to classify due to similarity with N class, but CNN still achieves 89% F1.
- 6. **Clinical Applicability** — High accuracy and reliable seizure detection make CNN suitable for real-world EEG analysis and epilepsy diagnosis support.
- 7. **LSTM Limitations** — Aggressive downsampling (4097→256 samples) likely caused information loss, limiting LSTM performance.
- 8. **Regularization Important** — L2 regularization, Dropout, and BatchNormalization helped prevent overfitting in both models.

